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Development Methodology

Matteo Zanioli

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Who I am

- Partner @Arsenalia
- Head of Technology (CTO) @Arsenalia
- Ph.D. in computer science @ Ca' Foscari University and École Normale Supérieure
- Executive MBA @ Bologna Business school
- Leading world wide projects about digital and ICT
- R&D on AI, IoT and blockchain

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What do we do in Arsenalia?

- cool projects ;-)

altitudo

Intelligent Aircraft Production with AI and Mixed Reality

Leonardo

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Who are you?

- A few simple questions, just to get to know each other. ☺

- https://app.wooclap.com/SYISBL

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What's your name

What is your name?

- Yeldana
- Faisal Akram
- f.g.
- Federico
- Marco
- Daniel Glazunov
- Santino
- Anna
- Giorgia
- Giacomo
- Stefano
- Deniz Bayan
- Nicola
- davide

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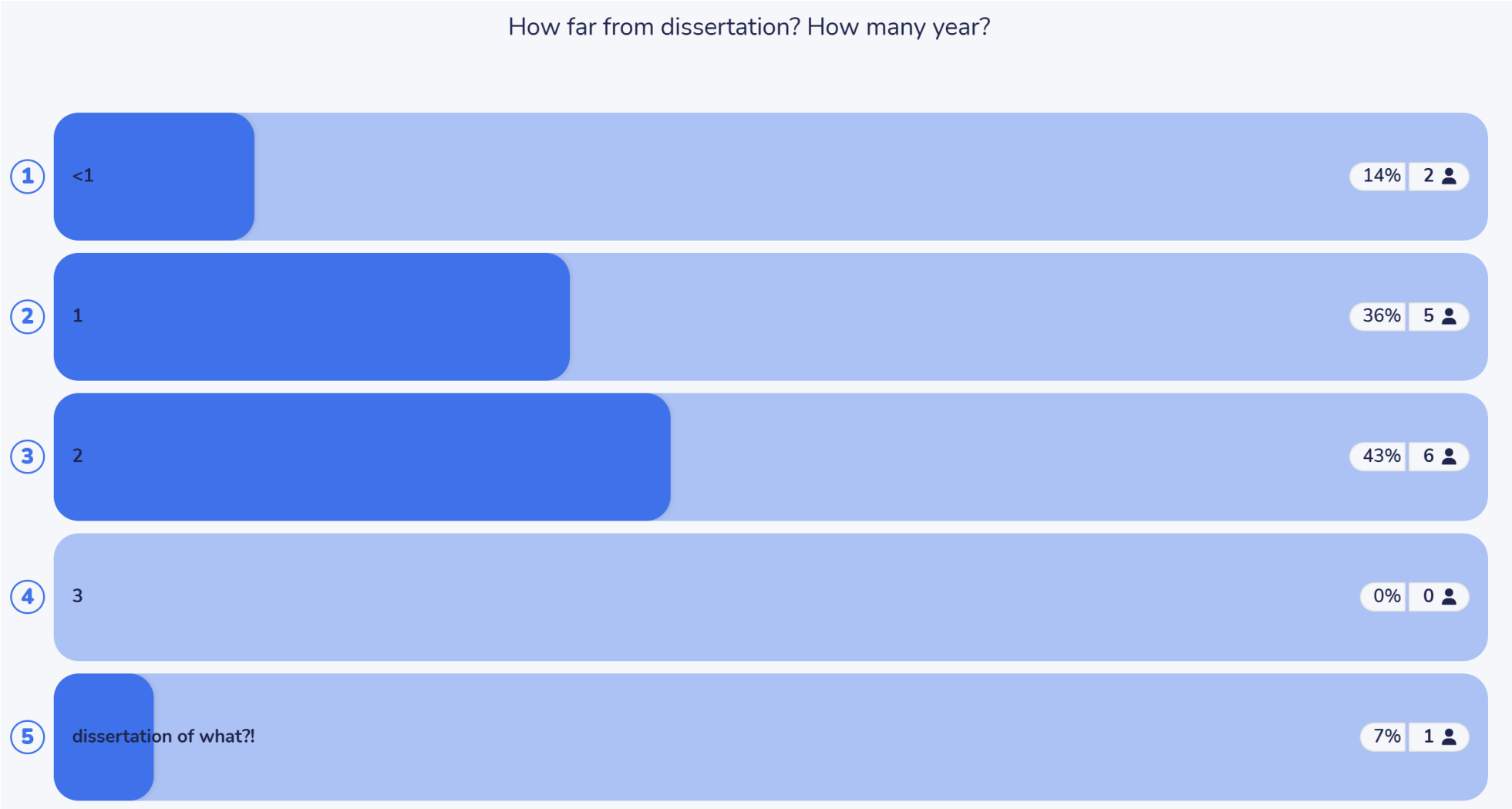
How far from dissertation



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Expectation from the course

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Expectation for your future

to degree and have a job

1

Working in a company

Hopefully graduate

Complete my degree and joining a Software company

to be autosufficient

to become a professor

to own wooclap

Lead engineer in a startup

just continuing with my job

Work in cybersecurity and teaching

Run my own software house

President

Use the knowledge for more artistic pursuits

working in a computer science company or doing research

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Best skills

Ability to find the solution by myself
2

alpinism
2

programming (mostly backend)
2

problem solving
2

Research
1

studying
1

working in a team,problem solver,teaching things

learning

Working in groups

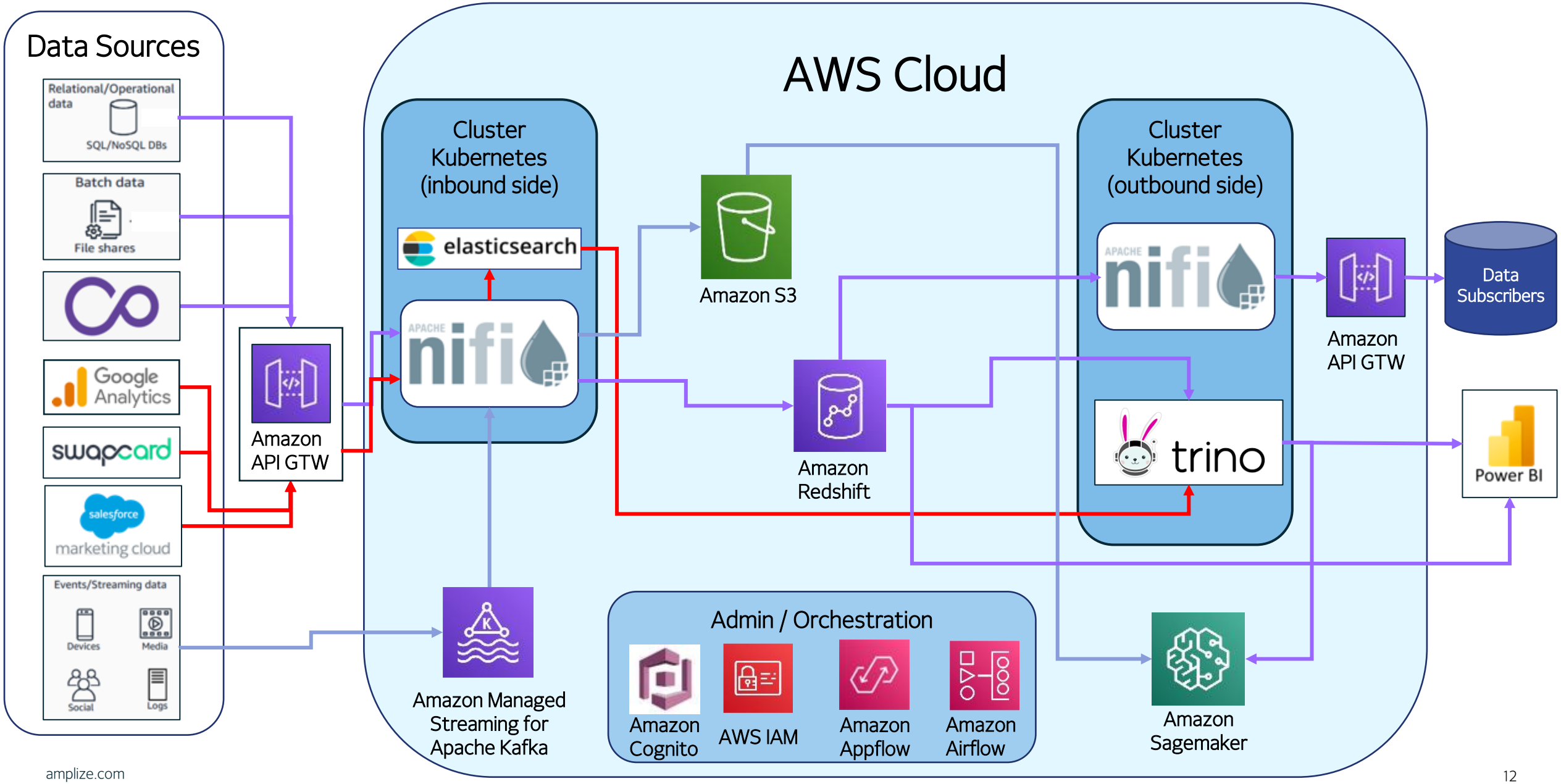
Problem Solver, Decision Maker

Sleeping 😴 Honk-mimimimimi

Let's start from the end…

Architecture

Arsenalia



key question

how we can handle complex projects with complex architecture?

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the importance of making the right choice

- The choice of the railway track gauge (1435mm) was based on the needs of the time. Now it poses limitations, such as for magnetic levitation (maglev) trains

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the importance of making the right choice

- One of the reason that we have wine bottle of 75cl is related to the capability of English wine importers to held two gallons of wine: it's equivalent to about 9 liters = 12 bottles of 75 cl.

the importance of making the right choice

- Frequently in computer science, a provisional fix or workaround ends up becoming the permanent solution.
- Many legacy systems remain operational due to the substantial expenses associated with refactoring.
- The way you approach your projects is crucial both for yourself and for anyone else who will need to collaborate on them, especially in the future.

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the main resource for learn the best methodology

- Experience
- Experience
- Experience

- There are multiple book, links, whitepaper (and so on) that describe different methodology: but not a single answer exists for each project

Content of the course

- Same as "Software Architecture" course, "Development Methodology" is a very practical topic: many solutions, solutions change in time and nobody knows all solutions.
- Please: Always think with your brain, considering requirements and constraints.
- The main goals of the course are to give you
  - some tools to understand different methodology and apply them
  - some base knowledge to setup a (big) project in the right way

What is Development Methodology?

What is Development Methodology?

- ChatGPT: "*Development methodology is a **structured approach** to software development that outlines **processes** and **practices** for **planning**, **creating**, and **delivering** software. It includes various models like Waterfall, Agile, Scrum, and DevOps, each offering different ways to manage and execute projects based on specific **needs** and **goals**.*"

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Teaching approach

- We could discuss methodology for months, but both you and I have many other commitments.
- I will endeavour to blend theory with practice to offer you as much support as possible.
- I can point you in the right direction and assist with your initial steps, but ultimately, the journey is yours to take alone.

Course outline

- I had defined all topics, but they can change during the course for different reasons:
  - Our speed (maybe I will be slower than expected, or you will be faster than expected)
  - Other topics might arise while teaching the lectures -> feel free to propose topics

Course outline

- Methodology for development complex application
- We will understand the use of deployment system and automated tests and all other tools
- I present you main risks that can rise on a project and how to avoid that (or at least reduce that)

- Note: this is not a Project Management course, we will not talk about time, costs, managing resources, etc.

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My Expectation

- Curiosity, willingness to learn, and dedication
  - I would like to engage with you on the topics during in-person lessons
  - I can also provide real-world examples from the industry, but I need your cooperation to gauge the appropriate level of detail to cover
- Mutual Respect
  - For me: if I am boring or I am discussing something trivial or unimportant to you, please raise your hand so I can better address the topic.
  - For you: if anything is unclear, feel free to raise your hand – everyone is here to learn.
  - For your classmates: kindly avoid causing disruptions during the lesson.

Textbook

- Let me know if you find something interesting ☺
- I would suggest these references
  - Joel on Software
  - Phoenix Project
  - https://refactory.fm
- I will share time to time link or other references related on the specific topics

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How communicate with me?

- For one-to-one topics, send me an e-mail: matteo.zanioli@unive.it
- Linkedin: https://www.linkedin.com/in/matteozanioli/
- Please, before rise a question try to find the answer: if you have found an answer, let's discuss it together!

Exam

- Written exam
- A case to manage and organize properly explaining your choice
- Evaluation elements
  - Soundness
  - Completeness
  - Conciseness
  - Attention on the details

Software Architecture + Development Methodologies

- Some of you need to take also the Software Architecture exams
  - Two modules of a 12 ECTS course
- In this case
  - You need to pass both the modules in the same academic year
    - i.e., if you pass one module but not the other one you'll need to retake the full exam next year
  - The final grade will be the average of the two grades
- The written exam of the two modules will be at the same time
  - Split in two sessions with a short break in the middle to allow people that take only one module to enter and exit the exam room

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Additional material

- We must provide additional material to some students
  - Working students
  - With difficulties in learning
  - See here https://www.unive.it/pag/42819/
- As additional material, those students will have access to the recording of my lectures in the Moodle space
  - Note that I cannot decide who I allowed to have access
  - In case you think you should have access, but you don't have it, contact directly Ca' Foscari offices

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Calendar

- Thursday 14:00-15:30 and Friday 15:45-17:15

- All lectures in presence in Laboratorio 3 for Thursday and Aula A for Friday – Campus scientifico via Torino (building Zeta)

Useful technologies

- GitHub
  - Premium account of GitHub for free for students and professors (https://education.github.com/benefits)
  - Github copilot
- AWS / Azure
  - AWS free tier (https://aws.amazon.com/free/)
  - Azure for students (https://azure.microsoft.com/en-us/free/students)
- Vscode / IDE / Cursor AI
  - https://code.visualstudio.com/
- Stackoverflow and AI (Gemini, ChatGPT, Anthropic, ...)
- And much more TBD

Useful technology

- Git – Git is a distributed version control system that tracks changes in source code during software development, allowing multiple developers to collaborate efficiently.
- IDE – Integrated Development Environment
  - you can build your own environment using script
  - for this course VSCode is enough
  - IDE is useful to work with Java, for example
  - IntellJ IDEA is suggested from Software Architecture course

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Useful technology

- Coursor AI - Cursor AI is an AI-powered code editor designed to enhance productivity by providing intelligent coding assistance. It integrates deeply with your codebase, offering features like real-time pair programming, code generation, debugging, and refactoring suggestions.
- Github copilot - is an AI-powered coding assistant that suggests code, explains functions, and helps debug directly in your editor. It adapts to your context, supports many languages, and boosts productivity by reducing repetitive tasks.

How to use them?

- Write the code with the editor (or IDE)
- Save it on code repository
- Deploy it on a AWS/Azure container

- Homework: try to configure them.

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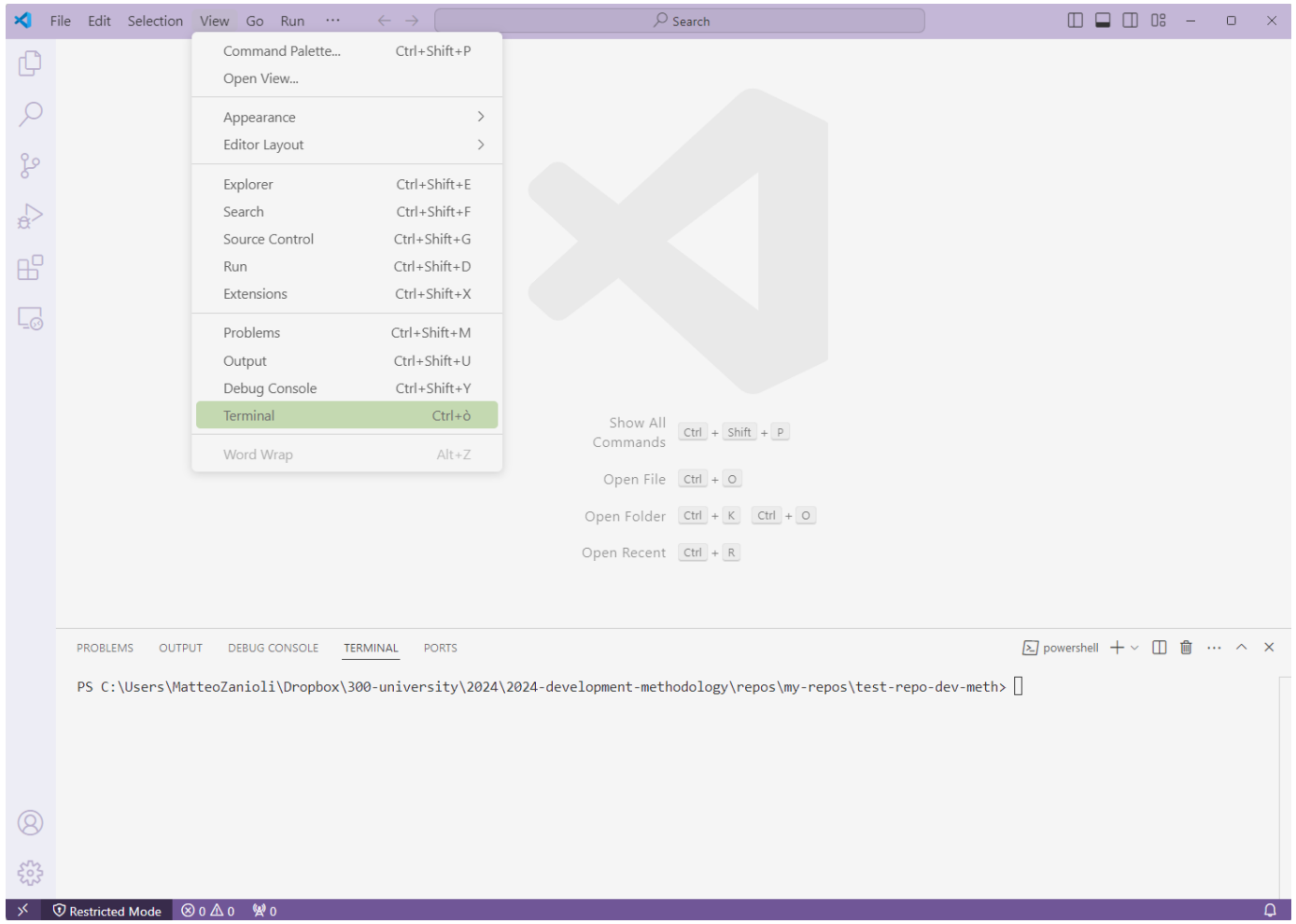
Environment configuration

- VSCode has a terminal
- Then you can launch your commands like

```
git clone <repo's address>
```

or

```
scp your-app.py <aws-space-to-run-your-application>
```



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Environment configuration

- You need to install git on your machine before to sync a repository
  - https://git-scm.com/doc
- If you need some support on that
  - ask to chatgpt or claude
  - try to look for it on google
  - only then ask to me ;-)
- For AWS you can register for free (actually they just take $1 from your credit card, in case you want buy some service)
  - a 5 pages form to fill
  - then you can create your private resources by the admin panel
  - depends on the resource you can proceed loading your code on the machine

Environment configuration

- For Azure you can register for free (with your unive account)
  - then you can create your private resources by the admin panel
  - depends on the resource you can proceed loading your code on the machine